

— AI ON YOUR TERMS

# A governed Claims Triage Agent for a UK insurer

From a raw claim submission to a routed, recorded, human-supervised decision — one agent workflow, built and hosted live.

Built on our own self-hosted n8n · live, reviewer-testable demo URL provided · OpenAI + PostgreSQL, served via Cloudflare

# Albion Mutual — a mid-size UK motor & home insurer

## WHO IS THE CUSTOMER

- UK general insurer, **~400k policies** across Motor, Home, Travel & Liability
- FCA-regulated; claims handled by a contact-centre plus a team of adjusters
- First Notice of Loss (FNOL) arrives by web form, email and phone

## THE PROBLEM THEY BROUGHT TO US

- Every new claim is read and triaged **by hand** before it reaches the right team
- High-value, fraud-suspect and vulnerable-customer cases are easy to miss in the queue
- Result: backlog, slow first response, inconsistent decisions, regulatory exposure

**~60%**

of triage effort is low-value, repetitive reading

**FCA**

Consumer Duty — must identify & protect vulnerable customers

**Now**

Claims volume up, headcount flat, scrutiny rising in 2026

# Triage is the highest-leverage place to put an agent

## CURRENT PAIN POINTS

- Manual classification & routing of every FNOL
- No consistent, auditable fraud-risk scoring at intake
- Vulnerable customers identified late, if at all
- Slow acknowledgement erodes customer trust

## EXPECTED BUSINESS VALUE

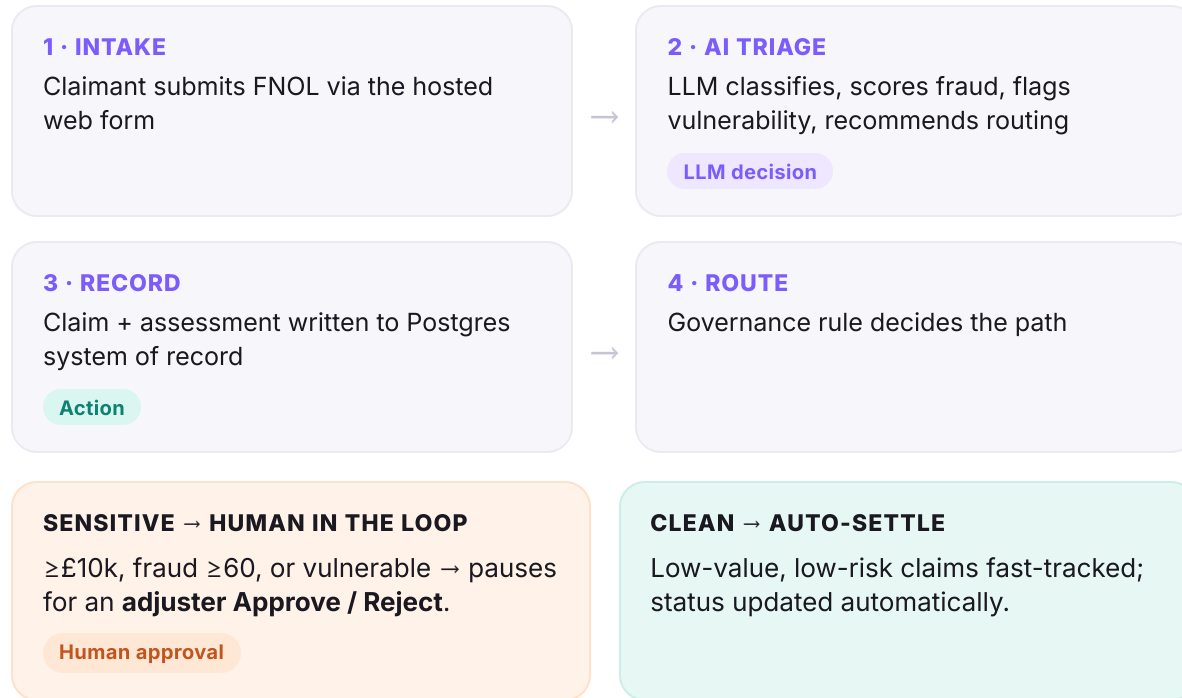
- Straight-through processing of clean, low-value claims
- Adjusters spend time on the claims that need judgement
- Consistent fraud & vulnerability flags from day one, fully audited
- Faster first response, fewer SLA breaches

■ ■ ■ Low-risk to pilot — humans stay in the loop on anything sensitive

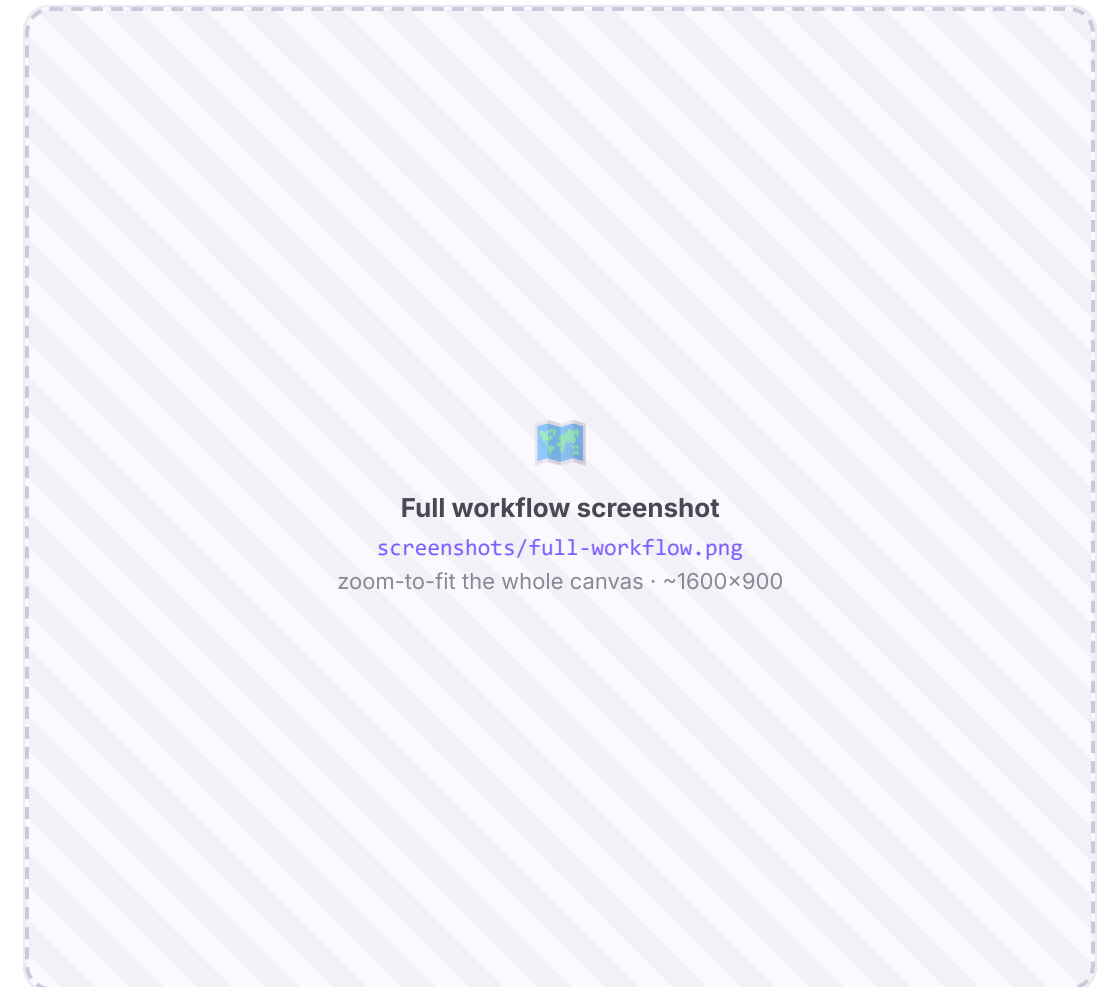
**Why prioritise it:** it touches a high-volume daily process, delivers measurable wins — speed, consistency, demonstrable Consumer-Duty compliance — within one quarter, and keeps a human accountable for every sensitive decision.



# End-to-end flow, orchestrated in n8n



The workflow on the n8n canvas



**Systems / data:** self-hosted **n8n** (our own production instance) · OpenAI GPT-4o-mini · PostgreSQL  
· Cloudflare tunnel for public hosting. **LLM role:** structured triage only — it never settles or rejects.  
**Human approval:** every high-value, high-fraud-risk or vulnerable case.

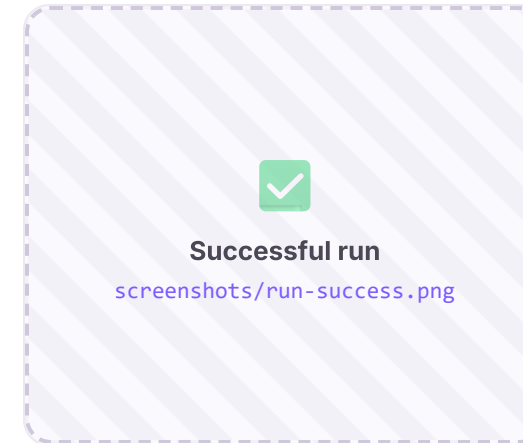
# Four claims, four correct outcomes

## WHAT RUNS LIVE

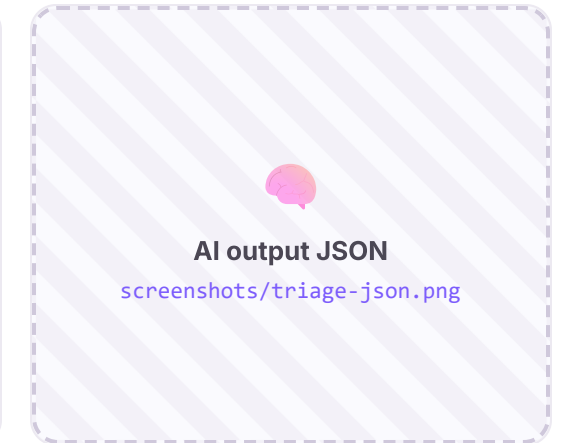
- **£480 motor scrape** → auto-settled, instant acknowledgement
- **£46k kitchen fire** → held for adjuster approval
- **Recent policy, round £9k theft, "settle this week"** → fraud 75, SIU review
- **Bereaved 78-year-old** → vulnerability flag, routed to a person

**Verified against the live model** — every route correct, fraud signals and the Consumer-Duty flag both fired as designed.

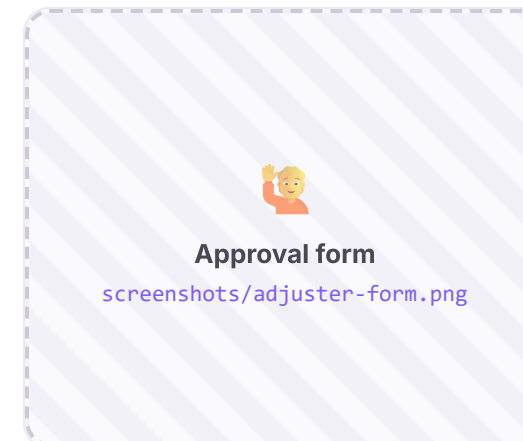
Successful run (green path)



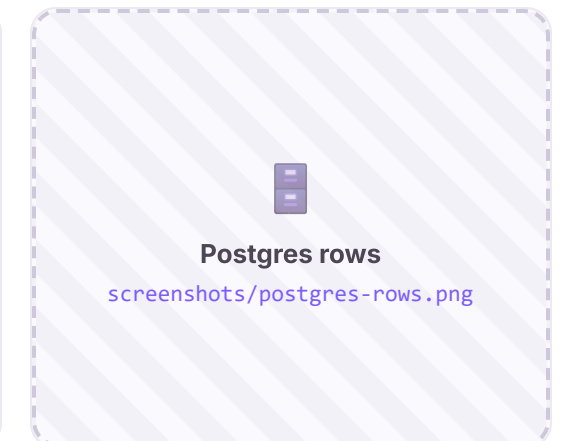
Structured AI triage output



Adjuster approval (human-in-loop)



Claims table in Postgres



# From a pilot to production-grade, governed AI

## WHY THIS IS THE JEEN STORY

- One flow orchestrates **data + model + human oversight** — not a chatbot bolted on the side
- The LLM is scoped to a decision; **actions and approvals are governed**
- Postgres gives a defensible **audit trail** for every claim and decision
- Human-in-the-loop is a **routing rule**, not an afterthought

## WHERE IT GOES NEXT

- Policy-system & fraud-database lookups as agent tools
- RAG over policy wordings for coverage checks
- Confidence thresholds & model fallback for resilience
- Dashboards on triage accuracy, STP rate, SLA & vulnerable-customer outcomes

**Deliverables:** exported n8n workflow (JSON) · full-workflow screenshot · successful-run screenshots · sample dataset · this deck · **plus a live hosted demo** reviewers can try.

# Built hands-on, on our own infrastructure

## HOW IT WAS BUILT & HOSTED

- Built on a **self-hosted, production n8n** instance we already run — not a trial or n8n Cloud
- Three workflows: the **FNOL agent** (form + human-in-the-loop), a **triage API** (webhook), and a **hosted portal** serving the UI
- Exposed over the public internet via our **Cloudflare tunnel** — reviewers get a live, clickable URL
- **Stack:** n8n · OpenAI GPT-4o-mini (structured output) · PostgreSQL (audit record) · Cloudflare
- Triage logic **validated against the live model** across 6 scenarios — every routing decision correct

■ Live demo: a reviewer-testable portal URL is included with this submission

## ASSIGNMENT REQUIREMENTS — ALL MET

- **Working agent flow** in n8n & fully demonstrable live
- **LLM-powered decision step** — structured triage (type, severity, fraud score, vulnerability, routing)
- **Action / integration** — Postgres write + email acknowledgement (not just chat)
- **Human-in-the-loop** — adjuster approval on every sensitive claim
- **Provided:** exported workflow JSON · full-flow screenshot · run screenshots · dataset
- **Presentation** — Customer/Problem · Why · Workflow overview